

#58 missing 2

$$\frac{9.5}{10}$$

Section 3.1 - 3.2

2 Let V = amount you could win

$V = 1$ for (T, T)

$V = 2$ for (H, H)

$V = -1$ for (H, T) or (T, H)

V	-1	1	2
Prob	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$

* 4 Let Y = # of open paths

$$P(\text{open}) = 6$$

$$Y = 0 \text{ for } (1 \cap 2 \cap 3)$$

$$Y = 1 \text{ for } (1 \cap 2' \cap 3') \dots (1' \cap 2 \cap 3)$$

$$Y = 2 \text{ for } (1' \cap 2' \cap 3')$$

Y	0	1	2
$P(Y)$	$\cancel{1/125}$	$\cancel{12/25}$	$64/125$

$$Y=0: (.2)(.8)(.2) + (.2)(.2)(.8) + (.2)(.2)(.2) = .072$$

$$Y=1: (.8)(.2)(.2) + (.8)(.2)(.8) + (.8)(.8)(.2) + (.2)(.8)(.8) = .416$$

6 Let Y = maximum of 2 sample picks

$$Y = 2 \text{ for } (1, 2)$$

$$Y = 3 \text{ for } (1, 3) \quad (2, 3)$$

$$Y = 4 \text{ for } (1, 4) \quad (2, 4) \quad (3, 4)$$

$$Y = 5 \text{ for } (1, 5) \quad (2, 5) \quad (3, 5) \quad (4, 5)$$

$$\text{Gotten from unique choices of } \binom{5}{2} = 10$$

$\therefore$ Each choice has a possibility of showing up in $1/10$ chance

Y	2	3	4	5
$P(Y)$	$1/10$	$1/5$	$3/10$	$2/5$

b. Let X = sum of 2 sample picks

$$X = 3 \text{ for } (1, 2)$$

$$X = 4 \text{ for } (1, 3)$$

$$X = 5 \text{ for } (1, 4), (2, 3)$$

$$X = 6 \text{ for } (1, 5), (2, 4)$$

$$X = 7 \text{ for } (2, 5), (3, 4)$$

$$X = 8 \text{ for } (3, 5)$$

$$X = 9 \text{ for } (4, 5)$$

Sample space still equal $\binom{5}{2} = 10$ since we pick without replacement

X	3	4	5	6	7	8	9
$P(X)$	$1/10$	$1/10$	$1/5$	$1/5$	$1/5$	$1/10$	$1/10$

1a Let Y = # number of errors detected, $P(\text{error}) = .05$

$Y = 0$ for zero errors in all 3 checks

$Y = 1$ for one errors in all 3 checks

$Y = 2$ for two

$Y = 3$ for all errors

$$P(Y=0) = (.95)^3 = \frac{6859}{8000}$$

$$P(Y=1) = \binom{3}{1} (.05) (.95)^2 = \frac{1083}{8000}$$

$$P(Y=2) = \binom{3}{2} (.05)^2 (.95) = \frac{57}{8000}$$

$$P(Y=3) = (.05)^3 = \frac{1}{8000}$$

Y	0	1	2	3
$P(Y)$	$\frac{6859}{8000}$	$\frac{1083}{8000}$	$\frac{57}{8000}$	$\frac{1}{8000}$

Section 3.3

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MA1 310

$$12 \quad E(Y) = \sum_{y=1}^4 y \cdot P(y) = (1 \times .4) + (2 \times .3) + (3 \times .2) + (4 \times .1) = 2$$

$$E(1/Y) = \sum_{y=1}^4 \frac{1}{y} \cdot P(y) = (1 \times .4) + (.5 \times .3) + (\frac{1}{3} \times .2) + (.25 \times .1) = \frac{77}{120}$$

$$c \quad E(Y^2 - 1) = E(Y^2) - E(1) = E(Y^2) - 1$$

$$E(Y^2) = \sum_{y=1}^4 y^2 P(y) = (1 \times .4) + (4 \times .3) + (9 \times .2) + (16 \times .1) = 5$$

$$E(Y^2 - 1) = 5 - 1 = 4$$

$$d. \quad V(Y) = E(Y^2) - [E(Y)]^2 = 5 - 2^2 = 1$$

$$20. \quad \text{Volume of 30 ft trailer} = 8 \times 10 \times 30 = 2400 \text{ ft}^3$$

$$\text{Volume of 40 ft trailer} = 8 \times 10 \times 40 = 3200 \text{ ft}^3$$

$$\begin{aligned} \text{Assignment} &= (P(30\text{ft}) \times V_{30\text{ft}}) + (P(40\text{ft}) \times V_{40\text{ft}}) \\ &= (.3 \times 2400) + (.7 \times 3200) \\ &= 2960 \end{aligned}$$

22

Let Y = # facing up on fair die

Y	1	2	3	4	5	6
$P(Y)$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$

$$E(Y) = \sum_{j=1}^6 Y \cdot P(Y) = (1 \cdot 1/6) + (2 \cdot 1/6) + (3 \cdot 1/6) + (4 \cdot 1/6) + (5 \cdot 1/6) + (6 \cdot 1/6) = 3.5$$

$$V(Y) = E(Y^2) - [E(Y)]^2$$

$$E(Y^2) = \sum_{j=1}^6 Y^2 \cdot P(Y) = (1 \cdot 1/6) + (4 \cdot 1/6) + (9 \cdot 1/6) + (16 \cdot 1/6) + (25 \cdot 1/6) + (36 \cdot 1/6) = 91/6$$

$$V(Y) = \frac{91}{6} - 3.5^2 = \frac{35}{12} \approx 2.9167$$

23

Let Y = amount received $Y = \$15$ for J or Q $Y = \$5$ for K or A $Y = -\$4$ for the rest

$$P(Y=15) = \binom{8}{1} \div \binom{52}{1} = \frac{2}{13}$$

$$P(Y=5) = \binom{8}{1} \div \binom{52}{1} = \frac{2}{13}$$

$$P(Y=-4) = \binom{(13-4) \times 4}{1} \div \binom{52}{1} = \frac{36}{52} = \frac{9}{13}$$

$$E(Y) = \sum_{j=1}^n Y \cdot P(Y) = (15 \times 2/13) + (5 \times 2/13) + (-4 \times 9/13) = \$4/13 \text{ or } \$0.3077 \text{ or } 31 \text{ cents}$$

33 Let Y be some discrete random variable with a, b as constants

$$\begin{aligned} a \quad E(aY + b) &= E(aY) + E(b) \\ &= a \cdot E(Y) + E(b) \\ &= a\mu + b \end{aligned}$$

b Prove $V(aY + b) = a^2 V(Y)$ Recall: $V(Y) = E[(Y - \mu)^2]$, where $\mu = E(Y)$ Let $Z = aY + b$ be a random variable

$$E(Z) = E(aY + b) = a\mu + b \text{ from (a) above}$$

$$\text{Let } m = a\mu + b$$

$$\begin{aligned} V(Z) &= E[(Z - m)^2] \\ &= E[(aY + b - (a\mu + b))^2] \end{aligned}$$

$$\begin{aligned}
 V(Z) &= E[(aY - a\mu)^2] \\
 &= E[a^2(Y - \mu)^2] \\
 &= a^2 E[(Y - \mu)^2] \\
 &= a^2 V(Y)
 \end{aligned}$$

MAT 3U

Nice!

34 Cost, $C = 10Y$

where $Y = \#$ of times used

$$E(Y) = \sum_{j=0}^2 j P(Y) = 0 + (1 \times 5) + (2 \times 4) = 1.3$$

Mean daily cost $= 10 \cdot 1.3 = \$13$

b. $V(Y) = E(Y^2) - [E(Y)]^2$

$$E(Y^2) = \sum_{j=0}^2 j^2 P(Y) = 0 + (1 \times 5) + (4 \times 4) = 2.1$$

$$V(Y) = 2.1 - [1.3]^2$$

$$= .41$$

From 23, we see that adding a constant to variance of a random variable has zero effect, while multiplying by a constant, \$10 in this case, increases variance by scale of 10^2 .
 So, $V(C) = V(10Y) = 10^2 V(Y) = 100 V(Y) = 100 \times .41 = \41

Section 3.1

70 $P(\text{recover}) = .6$

Let $Y = \#$ of people recover. $Y \sim \text{Bin}(20, .6)$

a. $P(Y=11) = \binom{20}{11} \cdot .6^{11} \cdot .2^9 = 1091$

b.
$$\begin{aligned} P(Y \geq 10) &= 1 - P(Y \leq 9) \\ &= 1 - \sum_{y=0}^9 \binom{20}{y} \cdot .6^y \cdot .2^{20-y} \\ &= 1 - 5.6341 \times 10^{-4} \\ &= .9994 \end{aligned}$$

c.
$$\begin{aligned} P(14 \leq Y \leq 16) &= \sum_{y=14}^{16} \binom{20}{y} \cdot .6^y \cdot .2^{20-y} \\ &= .8941 \end{aligned}$$

d.
$$\begin{aligned} P(Y \leq 16) &= \sum_{y=0}^{16} \binom{20}{y} \cdot .6^y \cdot .2^{20-y} \\ &= .5886 \end{aligned}$$

71. Let $Y = \#$ of graduate students. $Y \sim \text{Bin}(15, .2)$

$$\begin{aligned} P(Y \geq 10) &= 1 - P(Y \leq 9) \\ &= 1 - \sum_{y=0}^9 \binom{15}{y} \cdot .2^y \cdot .8^{15-y} \\ &= 1 - .9999 \\ &= .0001 \end{aligned}$$

43 $P(\text{rate}) = .7$

Let $Y = \#$ of qualified residents. $Y \sim \text{Bin}(5, .7)$

a.
$$\begin{aligned} P(Y=5) &= \binom{5}{5} \cdot .7^5 \\ &= .1681 \end{aligned}$$

$$\begin{aligned} b. P(Y \geq 4) &= P(Y=4) + P(Y=5) \\ P(Y=4) &= \binom{5}{4} \cdot .7^4 \cdot .3 = .36015 \\ P(Y \geq 4) &= P(Y=5) + P(Y=4) \\ &= .5282 \end{aligned}$$

56 Let $Y = \#$ of successful experiment. $Y \sim \text{Bin}(10, .1)$

$E(Y) = np = (10)(.1) = 1$

$V(Y) = np(1-p) = 1(1-.1) = .9$

→ this is not #58. This is #60.

58. Let $Y = \#$ of immune fish. $Y \sim \text{Bin}(20, .8)$

a. $P(Y=14) = \binom{20}{14} \cdot .8^{14} \cdot .2^6$
 $= .1091$

b. $P(Y \geq 10) = 1 - P(Y \leq 9)$
 $= 1 - \sum_{y=0}^9 \binom{20}{y} \cdot .8^y \cdot .2^{20-y}$
 $= 1 - .0006$
 $= .9994$

c. $P(Y \leq 16) = \sum_{y=0}^{16} \binom{20}{y} \cdot .8^y \cdot .2^{20-y}$
 $= .5886$

d. $E(Y) = np = (20)(.8) = 16$
 $V(Y) = np(1-p) = 16(1-.8) = 3.2$